

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 1 — Atomic Structure

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	1.1 Particles & Radii · 1.2 Isotopes & MS · 1.3 Orbitals · 1.4 Ionisation Energy
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	241 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

1.1 Particles in the atom and atomic radius

— Proton/neutron/electron mass and charge, electric field deflection, periodic and ionic radii

1.2 Isotopes and mass spectrometry

— Definitions, isotopic notation, mass spectra interpretation, Ar calculations, diatomic Cl₂ fragments

1.3 Electrons, energy levels and atomic orbitals

— Aufbau principle, s/p orbital geometry, full configurations, anomalous Cr/Cu, Hund's rule

1.4 Ionisation energy

— Definitions, equations, Period 3 general trends, Al and S anomalies, successive IE group deduction

SUBTOPIC 1.1 — PARTICLES IN THE ATOM & ATOMIC RADIUS

1 Sub-Atomic Particles — 9701/12/M/J/21/Q1

EASY • Syllabus 1.1

For each question, select the correct option.

- (a) Which statement about the particles present in an atom of ^{31}P is correct? [1]
 A The nucleus contains 15 protons and 16 neutrons.
 B The number of occupied electron shells is 4.
 C The mass of an electron is approximately $1/184$ of the mass of a proton.
 D The nucleus has an overall negative charge.
- (b) A beam of protons, neutrons, and electrons passes through a uniform electric field between two charged plates. Which particle is deflected by the greatest angle? [1]
 A Proton, because it has a positive charge.
 B Neutron, because it has no electrical charge.
 C Electron, because it has the smallest mass-to-charge ratio.
 D Both proton and electron are deflected by the same angle.

2 Fundamental Properties of Subatomic Particles — 9701/22/O/N/22/Q1(a)

EASY • Syllabus 1.1

Atoms are composed of three fundamental subatomic particles: protons, neutrons, and electrons.

- (a) Complete Table 1.1 to state the relative mass and relative charge of each particle. [3]

- (b) Explain why neutrons are not deflected when passed through an electric field. [1]

3 Atomic and Ionic Radii Comparison — 9701/21/M/J/22/Q1(b)

EASY • Syllabus 1.1

Sodium and chlorine are elements located in Period 3 of the Periodic Table.

- (a) Explain why the ionic radius of a sodium ion, Na^+ , is significantly smaller than the atomic radius of a sodium atom, Na . [2]

- (b) Explain why the ionic radius of a chloride ion, Cl^- , is larger than the atomic radius of a chlorine atom, Cl . [2]

4 Interpreting Periodic Trends in Atomic Radius — 9701/22/F/M/23/Q1(a)–(c)

HARD • Syllabus 1.1

Fig. 4.1 shows the variation in atomic radius across Period 3 of the Periodic Table from sodium to argon.

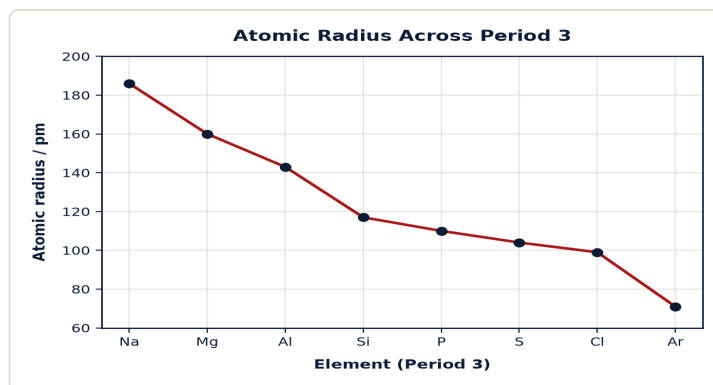


Fig. 4.1 Variation of atomic radius across Period 3 elements

- (a) Use Fig. 4.1 to state the atomic radius of silicon and the atomic radius of sulfur. [2]

.....

.....

- (b) Describe the general trend in atomic radius across Period 3 from Na to Cl. [1]

.....

.....

- (c) Explain this trend across Period 3 in terms of nuclear charge, shielding, and electron attraction. [3]

.....

.....

.....

.....

.....

5 Isoelectronic Species Comparison — 9701/23/M/J/23/Q1(c)

HARD · Syllabus 1.1

The nitride ion (N^{3-}), oxide ion (O^{2-}), fluoride ion (F^-), sodium ion (Na^+), and magnesium ion (Mg^{2+}) are isoelectronic species.

- (a) Define the term isoelectronic. [1]

.....

.....

- (b) Write the full electron configuration for the N^{3-} ion. [1]

.....

.....

- (c) Arrange the five ions in order of decreasing ionic radius (largest first) and explain your reasoning fully. [3]

.....

.....

.....

.....

.....

6 Behaviour of Charged Particles in Fields — 9701/22/O/N/21/Q1(b)

HARD · Syllabus 1.1

A collimated beam consisting of H^+ , 2H^+ , and 4He^{2+} ions passes at constant velocity through a uniform electric field between two horizontal parallel plates.

- (a) Deduce the direction of deflection of the ions. [1]

.....

.....

- (b) Calculate the charge-to-mass ratio (q/m) for each of the three ions: 1H^+ , 2H^+ , and 4He^{2+} . [3]

.....

.....

.....

- (c) State, with a reason, which of these ions will show identical deflection in the electric field. [2]

.....

.....

.....

SUBTOPIC 1.2 — ISOTOPES & MASS SPECTROMETRY

7 Defining and Representing Isotopes — 9701/21/M/J/21/Q1(a)

EASY · Syllabus 1.2

Bromine exists naturally as two stable isotopes, bromine-79 and bromine-81.

- (a) Define the term isotope. [2]

.....

.....

.....

- (b) Write the isotopic notation (including atomic number and mass number) for bromine-79 and bromine-81. [2]

.....

.....

- (c) Explain why bromine-79 and bromine-81 exhibit identical chemical reactivity. [2]

.....

.....

.....

8 Physical Properties of Isotopes — 9701/11/O/N/23/Q2

EASY · Syllabus 1.2

Hydrogen has three isotopes: protium (1H), deuterium (2H), and tritium (3H).

- (a) Which property is identical for pure H_2O and pure D_2O (where D is deuterium)? [1]

- A Density at 20°C
B Boiling point at 1 atm
C Empirical formula
D Relative molecular mass

- (b) Suggest why heavy water, D_2O , has a slightly higher boiling point than standard water, H_2O . [2]

.....

.....

.....

9 Relative Atomic Mass from Mass Spectrometry — 9701/22/M/J/22/Q1(c)

HARD · Syllabus 1.2

A naturally occurring sample of element Q was analysed using a mass spectrometer. The resulting spectrum is shown in Fig. 9.1.

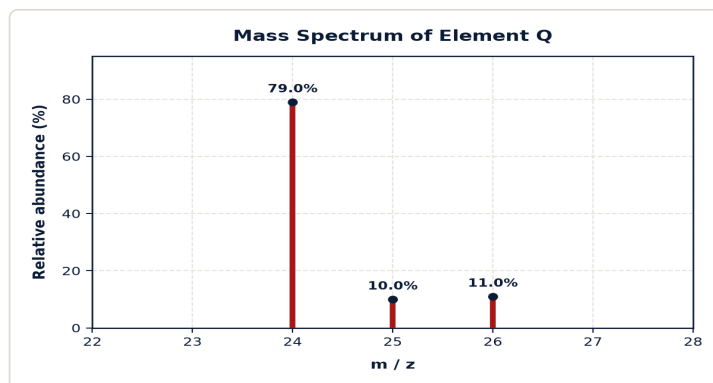


Fig. 9.1 Mass spectrum showing relative abundance of isotopes of element Q

- (a) State what is measured on the horizontal axis (m/z) of the mass spectrum. [1]

.....

.....

- (b) Calculate the relative atomic mass, A_r , of element Q to two decimal places using the data in Fig. 9.1. [3]

.....

.....

.....

- (c) Identify element Q from your calculated A_r value. [1]

.....

.....

10 Mass Spectrometry of Diatomic Molecules — 9701/21/O/N/22/Q1(b)–(c)

HARD · Syllabus 1.2

Chlorine exists as two isotopes: ^{35}Cl (relative abundance 75%) and ^{37}Cl (relative abundance 25%). A sample of pure chlorine gas, Cl_2 , is ionised in a mass spectrometer to form molecular ions, Cl_2^+ .

- (a) Predict the m/z values of the three molecular ion peaks observed in the mass spectrum of Cl_2 . [2]

.....

.....

.....

- (b) Calculate the expected ratio of the peak heights for the three molecular ions. [3]

.....

.....

.....

- (c) Explain why peaks are also detected at $m/z = 35$ and $m/z = 37$. [1]

.....

.....

11 Defining Relative Atomic Mass — 9701/22/F/M/21/Q1(b)

EASY · Syllabus 1.2

Standard atomic weights published in the IUPAC Periodic Table are determined relative to the carbon-12 standard.

- (a) Define the term relative atomic mass, A_r . [2]

.....

.....

.....

- (b) State why carbon-12 was chosen as the international standard for atomic mass measurement. [1]

.....

.....

12 Determining Isotopic Abundances Algebraically — 9701/23/M/J/24/Q1(a)

HARD · Syllabus 1.2

Gallium has a relative atomic mass of 69.72 and consists of two isotopes: ^{69}Ga and ^{71}Ga .

- (a) Let the percentage abundance of ^{69}Ga be $x\%$. Express the percentage abundance of ^{71}Ga in terms of x . [1]

.....

.....

- (b) Set up an equation and calculate the percentage abundance of ^{69}Ga and ^{71}Ga in the naturally occurring sample. Show your working. [3]

.....

.....

.....

.....

.....

SUBTOPIC 1.3 — ELECTRONS, ENERGY LEVELS & ATOMIC ORBITALS

13 Atomic Orbitals and Quantum Numbers — 9701/12/M/J/22/Q3

EASY · Syllabus 1.3

Electrons occupy orbitals within sub-shells and shells designated by principal quantum number n .

- (a) What is the maximum number of electrons that can occupy a 3d sub-shell? [1]
- A 2
- B 6
- C 10
- D 14

- (b) How many total orbitals are present in the principal quantum shell $n = 3$? [1]
- A 3
- B 4
- C 9
- D 18

- (c) State the three-dimensional geometric shape of an s orbital and of a p_x orbital. [2]

.....

.....

.....

14 Sub-Shell Energy Ordering and Aufbau Principle — 9701/22/O/N/23/Q1(b)

EASY · Syllabus 1.3

Fig. 14.1 illustrates the relative energy levels of atomic orbitals from 1s to 3d.

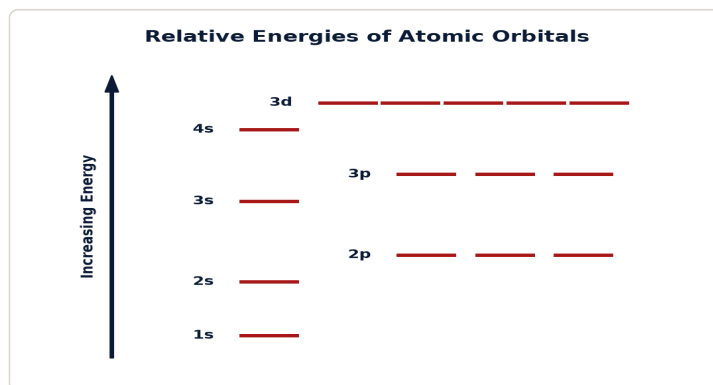


Fig. 14.1 Relative energies of atomic orbitals up to 3d

- (a) Using Fig. 14.1, state which sub-shell has a lower energy: 4s or 3d. [1]
-
-
- (b) Explain why the 4s sub-shell fills before the 3d sub-shell when constructing the electron configuration of calcium ($Z = 20$). [2]
-
-
-
- (c) Explain why 4s electrons are removed before 3d electrons when a transition metal atom forms an ion. [2]
-
-
-

15 Full and Shorthand Configurations of Atoms and Ions — 9701/21/M/J/23/Q1(a)–(b)

HARD · Syllabus 1.3

Write full or noble gas core electron configurations for the following species.

- (a) Write the full 1s... electron configuration of a sulfur atom, S ($Z = 16$). [1]
-
-
- (b) Write the full electron configuration of a copper atom, Cu ($Z = 29$). [1]
-
-
-
- (c) Explain why the electronic configuration of copper is considered anomalous compared to earlier Period 4 d-block elements. [2]
-
-
-
- (d) Write the shorthand (noble gas core) configuration of the iron(III) ion, Fe^{3+} ($Z = 26$). [2]
-
-
-

16 Electrons in Boxes and Hund's Rule — 9701/22/F/M/22/Q1(c)

HARD · Syllabus 1.3

Hund's rule and the Pauli exclusion principle govern how electrons populate degenerate orbitals.

- (a) State Hund's rule of maximum multiplicity. [2]

.....

.....

.....

- (b) Describe an electrons-in-boxes representation of the 2p sub-shell of a nitrogen atom ($Z = 7$). [2]

.....

.....

.....

- (c) Describe an electrons-in-boxes representation of the 3d sub-shell of a Mn^{2+} ion ($Z = 25$). State the number of unpaired electrons. [2]

.....

.....

.....

17 Free Radicals and Unpaired Electrons — 9701/13/O/N/21/Q4

EASY · Syllabus 1.3

A free radical is an atom, ion, or molecule containing at least one unpaired valence electron.

- (a) Which species is classified as a free radical? [1]

- A Cl^-
B $Cl^\bullet$
C Cl_2
D HCl

- (b) State the total number of unpaired electrons in a neutral phosphorus atom ($Z = 15$). [1]

.....

.....

- (c) Explain, in terms of electronic configuration, why the chloride ion, Cl^- , is chemically unreactive compared to a chlorine free radical, $Cl^\bullet$. [2]

.....

.....

.....

18 Predicting Group and Period from Configuration — 9701/21/M/J/24/Q1(d)

EASY · Syllabus 1.3

An unknown element X forms an X^{2-} ion with the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6$.

- (a) Deduce the electron configuration of a neutral atom of element X. [1]

.....

.....

- (b) State the group and period of element X in the Periodic Table. [2]

.....

.....

- (c) Identify element X.

[1]

SUBTOPIC 1.4 — IONISATION ENERGY

19 Definitions and Equations for Ionisation Energy — 9701/22/M/J/21/Q2(a)

EASY · Syllabus 1.4

Ionisation energy is a fundamental property reflecting the electronic structure of atoms.

- (a) Define the term first ionisation energy.

[3]

- (b) Write an equation, including state symbols, representing the second ionisation energy of magnesium.

[2]

20 Period 3 First Ionisation Energy Trend and Anomalies — 9701/22/F/M/22/Q2(a)–(d)

HARD · Syllabus 1.4

Fig. 20.1 shows the variation in first ionisation energy across the elements of Period 3 from sodium to argon.

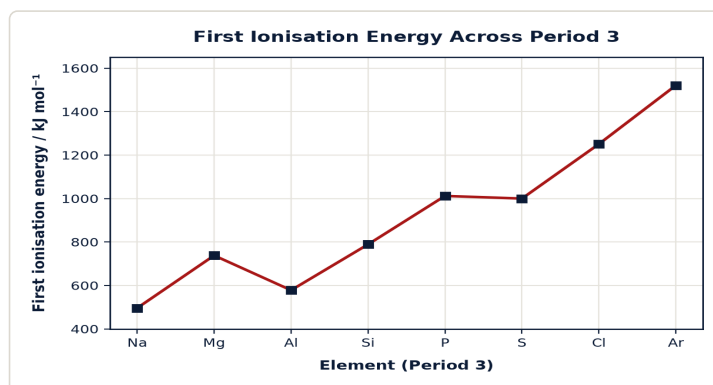


Fig. 20.1 First ionisation energies across Period 3 elements

- (a) State and explain the general trend in first ionisation energy across Period 3 from Na to Ar.

[3]

- (b) Explain why the first ionisation energy of aluminium is lower than that of magnesium, despite aluminium having a greater nuclear charge.

[3]

- (c) Explain why the first ionisation energy of sulfur is lower than that of phosphorus.

[3]

.....

.....

.....

.....

21 **Deducing Group from Successive Ionisation Energies — 9701/21/O/N/23/Q1(c)–(e)**

HARD · Syllabus 1.4

Fig. 21.1 displays the successive ionisation energies (as $\log_{10}$ IE) for all 15 electrons of an unknown element X.

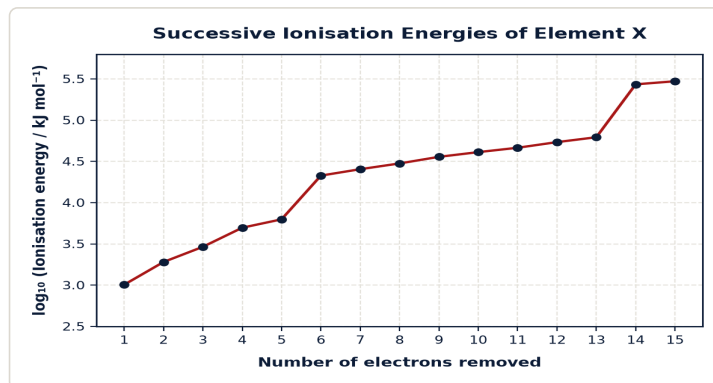


Fig. 21.1 Plot of $\log_{10}(\text{ionisation energy})$ against electron number for element X

- (a) Explain why successive ionisation energies always increase for any given element.

[2]

.....

.....

.....

- (b) Identify the electron numbers between which the largest jumps in ionisation energy occur in Fig. 21.1.

[2]

.....

.....

.....

- (c) Deduce the group of the Periodic Table to which element X belongs. Justify your answer using the graph.

[2]

.....

.....

.....

- (d) Identify element X.

[1]

.....

.....

22 **Successive Ionisation Energy Data Analysis — 9701/22/M/J/24/Q1(b)**

EASY · Syllabus 1.4

The first six successive ionisation energies of an element Y (in kJ mol^{-1}) are: 578, 1817, 2745, 11578, 14831, 18378.

- (a) Identify between which two ionisations the largest relative increase occurs.

[1]

.....

.....

- (b) State the number of valence electrons present in an atom of element Y. [1]

.....

.....

- (c) Deduce the formula of the chloride formed by element Y. [1]

.....

.....

23 Down Group Ionisation Energy Trends — 9701/11/M/J/23/Q4

EASY · Syllabus 1.4

Consider the elements of Group 2: beryllium, magnesium, calcium, strontium, and barium.

- (a) Describe the trend in first ionisation energy down Group 2. [1]

.....

.....

- (b) Explain this trend in terms of atomic radius and electron shielding, despite the increasing nuclear charge. [3]

.....

.....

.....

.....

24 Extended Evaluative Question: Periodic Trends — 9701/22/O/N/24/Q1(e)

HARD · Syllabus 1.4

'First ionisation energy increases monotonically across Period 3 without exception because nuclear charge increases by +1 proton at each successive element.'

- (a) Evaluate this statement. In your answer: [6]

- State whether the statement is completely true, partially true, or false.
- Explain the general trend across the period.
- Identify and explain the two specific exceptions across Period 3, citing electronic configurations.

.....

.....

.....

.....

.....

.....

.....

.....

SUBTOPIC 1.1 — PARTICLES IN THE ATOM & ATOMIC RADIUS

25 Deflection of Particles in Electric Fields — 9701/22/F/M/24/Q1(a)-(c)

HARD · Syllabus 1.1

Fig. 25.1 shows the paths of three different beams of subatomic particles passing through a uniform electric field between two charged parallel plates.

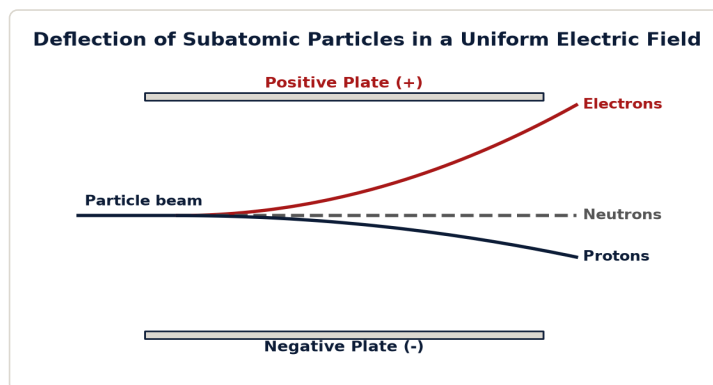


Fig. 25.1 Deflection of subatomic particles in a uniform electric field

- (a) State the identity of the particles responsible for each of the three paths shown in Fig. 25.1. [3]

.....

.....

.....

- (b) Explain why electrons are deflected by a significantly greater angle than protons, even though both carry the same magnitude of electric charge. [2]

.....

.....

.....

- (c) State two factors that determine the angle of deflection of a charged particle in a uniform electric field. [2]

.....

.....

.....

26 Deducing Subatomic Particle Numbers in Ions — 9701/12/M/J/23/Q1

EASY · Syllabus 1.1

Table 26.1 lists four different ionic species.

- (a) Deduce the number of protons, neutrons, and electrons present in one $^{34}\text{S}^{2-}$ ion. [3]

.....

.....

.....

- (b) Deduce the number of protons, neutrons, and electrons present in one $^{56}\text{Fe}^{3+}$ ion. [3]

.....

.....

.....

- (c) State which two of the following ions contain the same number of neutrons: $^{39}\text{K}^+$, $^{40}\text{Ca}^{2+}$, $^{35}\text{Cl}^-$, $^{37}\text{Cl}^-$. [1]

.....

.....

27 Trends in Halogen and Halide Radii — 9701/21/O/N/23/Q1(b)

EASY · Syllabus 1.1

The Group 17 elements (halogens) form halide ions with a 1^- charge.

- (a) Describe and explain the trend in atomic radius down Group 17 from fluorine to iodine. [3]

.....

.....

.....

.....

- (b) Explain why the ionic radius of a bromide ion, Br^- , is greater than the covalent radius of a bromine atom, Br. [2]

.....

.....

.....

28 Specific Charge Calculations — 9701/22/M/J/20/Q1(b)

HARD · Syllabus 1.1

The specific charge of a particle is defined as the ratio of its electric charge to its mass.

- (a) Calculate the specific charge (in C kg^{-1}) of a proton. (Elementary charge $e = 1.60 \times 10^{-19} \text{ C}$; proton mass $m_p = 1.67 \times 10^{-27} \text{ kg}$). [2]

.....

.....

.....

- (b) Calculate the specific charge of an alpha particle (4He^{2+}). [2]

.....

.....

.....

- (c) State the ratio of the specific charge of an electron to that of a proton. [1]

.....

.....

29 Effective Nuclear Charge Across Period 3 — 9701/23/M/J/21/Q1(a)

HARD · Syllabus 1.1

Effective nuclear charge (Z_{eff}) is the net positive charge experienced by valence electrons, taking into account shielding by core electrons.

- (a) Explain why the core electron shielding remains approximately constant across Period 3 from Na to Cl. [2]

.....

.....

.....

- (b) Describe how effective nuclear charge (Z_{eff}) changes from Na to Cl, and relate this to the decrease in atomic radius. [3]

.....

.....

.....

.....

30 Rutherford Alpha-Scattering and Nuclear Model — 9701/21/M/J/19/Q1(a)

EASY · Syllabus 1.1

In the Geiger-Marsden experiment, a thin gold foil was bombarded with alpha particles.

- (a) State the observation from the experiment that showed that most of the atom is empty space. [1]

.....

.....

- (b) State the observation that proved the existence of a small, dense, positively charged nucleus. [2]

.....

.....

.....

SUBTOPIC 1.2 — ISOTOPES & MASS SPECTROMETRY

31 High-Resolution Mass Spectrum of Zirconium — 9701/22/O/N/21/Q2(a)

HARD · Syllabus 1.2

Fig. 31.1 displays the mass spectrum of a naturally occurring sample of zirconium.

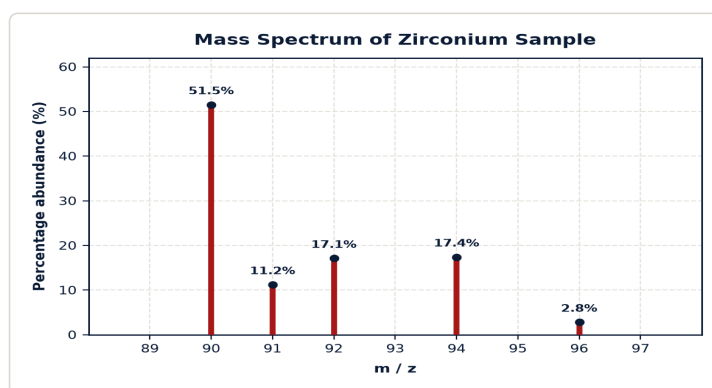


Fig. 31.1 Mass spectrum of a naturally occurring zirconium sample

- (a) State the number of stable isotopes of zirconium indicated by Fig. 31.1. [1]

.....

.....

- (b) Using the isotopic abundance data from Fig. 31.1, calculate the relative atomic mass (A_r) of zirconium to two decimal places. [3]

.....

.....

.....

- (c) State the number of neutrons present in the most abundant isotope of zirconium (atomic number $Z = 40$). [1]

.....

.....

32 Algebraic Determination of Boron Isotopes — 9701/11/F/M/22/Q2

EASY · Syllabus 1.2

Naturally occurring boron has a relative atomic mass of 10.81 and consists exclusively of two isotopes: ^{10}B and ^{11}B .

- (a) Which row gives the correct percentage abundance of ^{10}B and ^{11}B in natural boron? [1]
- A $^{10}\text{B} = 10.0\%$, $^{11}\text{B} = 90.0\%$
 B $^{10}\text{B} = 19.0\%$, $^{11}\text{B} = 81.0\%$
 C $^{10}\text{B} = 50.0\%$, $^{11}\text{B} = 50.0\%$
 D $^{10}\text{B} = 81.0\%$, $^{11}\text{B} = 19.0\%$

- (b) Show how the correct percentage abundances in part (a) can be calculated mathematically. [2]

.....

.....

.....

33 Mass Spectrometry of Lead Isotopes — 9701/22/M/J/23/Q1(a)

HARD · Syllabus 1.2

A sample of lead extracted from radioactive galena ore contains the four stable isotopes ^{204}Pb , ^{206}Pb , ^{207}Pb , and ^{208}Pb in relative abundances 1.4%, 24.1%, 22.1%, and 52.4% respectively.

- (a) Define the term relative isotopic mass. [2]

.....

.....

.....

- (b) Calculate the relative atomic mass of this sample of lead to one decimal place. [3]

.....

.....

.....

.....

34 Diatomic Bromine Molecular Ion Peak Ratios — 9701/21/O/N/24/Q1(b)

HARD · Syllabus 1.2

Bromine consists of ^{79}Br and ^{81}Br in an exact 1:1 abundance ratio. When gaseous bromine, Br_2 , is introduced into a mass spectrometer, molecular ions $[\text{Br}_2]^+$ are formed.

- (a) List the m/z values of the three molecular ion peaks observed in the mass spectrum of Br_2 . [2]

.....

.....

- (b) Deduce the ratio of the heights of the three peaks identified in part (a), showing your probability calculation. [3]

.....

.....

.....

.....

35 The Carbon-12 Standard and Unified Atomic Mass — 9701/23/M/J/22/Q1(c)

EASY · Syllabus 1.2

All atomic mass measurements in modern chemistry are calibrated against the unified atomic mass standard.

- (a) State the exact mass assigned by definition to one atom of carbon-12. [1]

.....

.....

- (b) Explain why the relative atomic mass of chlorine is 35.5 and not an integer, whereas the mass number of any individual isotope is always an integer. [2]

.....

.....

.....

36 Principles of Time-of-Flight (TOF) Mass Spectrometry — 9701/22/F/M/25/Q1(a)

EASY · Syllabus 1.2

In a time-of-flight (TOF) mass spectrometer, particles undergo four fundamental stages.

- (a) Name the four successive stages in TOF mass spectrometry. [2]

.....

.....

.....

- (b) Explain why all ions are given the same kinetic energy during the acceleration stage. [2]

.....

.....

.....

- (c) Deduce how the flight time of an ion in the drift region relates to its mass-to-charge ratio (m/z). [2]

.....

.....

.....

37 Determining Unknown Isotopic Mass from Peak Heights — 9701/21/M/J/25/Q1(c)

HARD · Syllabus 1.2

Silicon has $A_r = 28.09$. Its mass spectrum shows three peaks: ^{28}Si (92.2%), ^{29}Si (4.7%), and an unknown isotope $A\text{Si}$ (3.1%).

- (a) Calculate the mass number A of the third isotope of silicon. Show your working clearly. [3]

.....

.....

.....

.....

- (b) Deduce the number of neutrons in an atom of this third isotope (silicon atomic number = 14). [1]

.....

.....

SUBTOPIC 1.3 — ELECTRONS, ENERGY LEVELS & ATOMIC ORBITALS

38 Chromium and Copper Anomalous Configurations — 9701/22/M/J/20/Q2(a)

HARD · Syllabus 1.3

Chromium ($Z = 24$) and copper ($Z = 29$) do not follow the standard Aufbau filling pattern across the first transition series.

- (a) Write the full electronic configuration ($1s^2 \dots$) of a neutral chromium atom. [1]

.....

.....

- (b) Explain why the $[\text{Ar}] 3d^5 4s^1$ configuration of chromium is energetically favoured over $[\text{Ar}] 3d^4 4s^2$. [2]
-
-
-

- (c) Write the full electronic configuration of a copper(I) ion, Cu^+ . [1]
-
-

39 Electron Configurations of Transition Metal Ions — 9701/21/O/N/20/Q2(b)

EASY · Syllabus 1.3

When d-block transition metals form positive ions, electrons are lost in a specific order.

- (a) Write the shorthand noble gas core configuration for the Co^{2+} ion ($Z = 27$). [1]
-
-
- (b) Write the shorthand configuration for the Ni^{2+} ion ($Z = 28$). [1]
-
-
- (c) Explain why the 4s electrons are removed before 3d electrons when forming transition metal cations. [2]
-
-
-

40 Shell Capacities and Orbital Sub-Shells — 9701/12/O/N/22/Q3

EASY · Syllabus 1.3

The quantum mechanical model defines orbitals by principal quantum number n and orbital quantum type.

- (a) What is the total maximum number of electrons that can be held in the fourth principal quantum shell ($n = 4$)? [1]
- A 16
- B 18
- C 32
- D 50
- (b) State the names of all four sub-shells present in the $n = 4$ shell, and state the number of orbitals in each. [2]
-
-
-

41 Orbital Geometry and Degeneracy — 9701/22/F/M/23/Q2(a)

HARD · Syllabus 1.3

Electrons reside in three-dimensional probability distributions termed atomic orbitals.

- (a) Explain the meaning of the term degenerate orbitals. [1]
-
-

- (b) Describe the spatial orientation of the three degenerate 2p orbitals: 2p_x, 2p_y, and 2p_z. [2]

.....

.....

.....

- (c) State two differences between a 1s orbital and a 2s orbital. [2]

.....

.....

.....

42 Electron Configurations of Anions — 9701/21/M/J/22/Q2(b)

EASY · Syllabus 1.3

Non-metal atoms gain electrons to achieve stable noble-gas valence electron configurations.

- (a) Write the full electron configuration of a phosphide ion, P³⁻. [1]

.....

.....

- (b) State which noble gas has the identical electron configuration to a P³⁻ ion. [1]

.....

.....

- (c) Write the electron configuration of an oxide ion, O²⁻. [1]

.....

.....

43 Pauli Exclusion Principle and Orbital Diagrams — 9701/23/M/J/24/Q2(a)

HARD · Syllabus 1.3

The Pauli exclusion principle states that an orbital can hold a maximum of two electrons, provided they have opposite spins.

- (a) Explain why two electrons occupying the same orbital must possess opposite spins. [2]

.....

.....

.....

- (b) Describe an electrons-in-boxes diagram for an oxygen atom (Z = 8), indicating the number of paired and unpaired electrons in the 2p sub-shell. [3]

.....

.....

.....

.....

SUBTOPIC 1.4 — IONISATION ENERGY

44 Comparing Period 2 and Period 3 First Ionisation Energies — 9701/22/M/J/23/Q2(a)–(c)

HARD · Syllabus 1.4

Fig. 44.1 compares the first ionisation energies of the elements across Period 2 (Li to Ne) and Period 3 (Na to Ar).

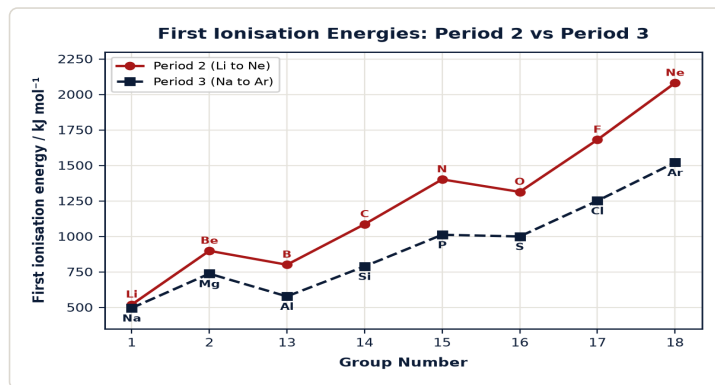


Fig. 44.1 Comparison of first ionisation energies across Period 2 and Period 3

- (a) State two general similarities in the pattern of first ionisation energies across Period 2 and Period 3 shown in Fig. 44.1. [2]
-
-
-
- (b) Explain why every Period 3 element has a significantly lower first ionisation energy than its corresponding Group counterpart in Period 2. [3]
-
-
-
-
- (c) Identify the two pairs of elements that display anomalies (dips) in both Period 2 and Period 3. [2]
-
-
-

45 Nitrogen vs Oxygen Ionisation Energy Anomaly — 9701/21/O/N/21/Q2(a)

HARD · Syllabus 1.4

The first ionisation energy of nitrogen is 1402 kJ mol⁻¹, whereas that of oxygen is lower at 1314 kJ mol⁻¹, despite oxygen having a greater nuclear charge (+8 vs +7).

- (a) Write the electronic configuration of a nitrogen atom and of an oxygen atom in 2p orbital notation. [2]
-
-
-
- (b) Explain why the first ionisation energy of oxygen is lower than that of nitrogen. [3]
-
-
-
-

46 Successive Ionisation Energies of Sodium — 9701/22/O/N/22/Q2(c)

HARD · Syllabus 1.4

The 11 successive ionisation energies of sodium (in kJ mol⁻¹) are: 496, 4563, 6913, 9544, 13352, 16611, 20115, 25491, 28934, 141367, 159079.

- (a) Identify between which two successive ionisations the first very large jump occurs. [1]
-
-

- (b) Explain why the 2nd ionisation energy of sodium is nearly 10 times greater than its 1st ionisation energy. [2]

.....

.....

.....

- (c) Explain why there is an immense jump between the 9th and 10th ionisation energies. [2]

.....

.....

.....

47 Deducing Identity from Successive IE Table — 9701/22/F/M/24/Q2(a)

EASY · Syllabus 1.4

An element Z in Period 2 of the Periodic Table exhibits the following first six ionisation energies: 1086, 2353, 4621, 6223, 37832, 47278 kJ mol⁻¹.

- (a) Identify the group of the Periodic Table to which element Z belongs. Justify your answer. [2]

.....

.....

.....

- (b) Deduce the chemical identity of element Z. [1]

.....

.....

- (c) Write the formula of the oxide formed by element Z in which Z displays its maximum oxidation state. [1]

.....

.....

48 Ionisation Energy Trends Down Group 1 — 9701/11/M/J/24/Q4

EASY · Syllabus 1.4

The first ionisation energies of Group 1 metals (alkali metals) are: Li (520), Na (496), K (419), Rb (403), Cs (376 kJ mol⁻¹).

- (a) Which factor is primarily responsible for the decrease in first ionisation energy down Group 1? [1]

- A Decreasing nuclear charge
B Increased atomic radius and shielding by inner electron shells
C Decreasing metallic bond strength
D Decreasing proton-to-neutron ratio

- (b) Explain why the increase in nuclear charge down Group 1 does not lead to an increase in first ionisation energy. [2]

.....

.....

.....

49 Thermochemical Equations for Ionisation Energies — 9701/21/M/J/21/Q2(a)

EASY · Syllabus 1.4

Equations representing ionisation energies must be balanced and include correct state symbols.

- (a) Write the equation representing the first ionisation energy of aluminium. [1]

.....

.....

- (b) Write the equation representing the third ionisation energy of aluminium. [2]

.....

.....

- (c) Explain why the 3rd ionisation energy of aluminium (2745 kJ mol^{-1}) is significantly higher than its 1st ionisation energy (578 kJ mol^{-1}). [2]

.....

.....

.....

50

Synoptic Problem: Atomic Structure & Ionisation Energies — 9701/22/O/N/25/Q1

HARD · Syllabus 1.4

Element M is a reactive metal. Its mass spectrum indicates a single isotope with nucleon number 24. Its first three ionisation energies are 738, 1451, and 7733 kJ mol^{-1} .

- (a) Deduce the group number of element M from its ionisation energy data, and write the full electron configuration of element M. [2]

.....

.....

.....

- (b) State the number of protons, neutrons, and electrons in an M^{2+} ion. [3]

.....

.....

.....

- (c) Explain, with reference to energy levels and sub-shells, why the jump between the 2nd and 3rd ionisation energies is so large. [3]

.....

.....

.....

.....

Mark Scheme — Topic 1: Atomic Structure

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	A — Phosphorus has atomic number 15 (15 protons) and nucleon number 31 (31 – 15 = 16 neutrons) (1)	1
1(b)	C — Deflection angle is proportional to charge/mass ratio; the electron has a much smaller mass than the proton (1)	1
2(a)	Proton: mass 1, charge +1 (1); Neutron: mass 1, charge 0 (1); Electron: mass 1/1836 to 1/1840 (or negligible), charge -1 (1)	3
2(b)	Neutrons have no charge / are electrically neutral, so they experience no electrostatic force (1)	1
3(a)	Na ⁺ has lost its outermost (3s) electron shell / has one fewer electron shell than Na (1); the remaining electrons experience a greater effective nuclear attraction / greater proton-to-electron ratio (1)	2
3(b)	Cl ⁻ gains an electron into the 3p sub-shell, causing increased electron-electron repulsion (1); while the nuclear charge remains constant (+17), expanding the electron cloud (1)	2
4(a)	Silicon: 117 pm (±2 pm) (1); Sulfur: 104 pm (±2 pm) (1)	2
4(b)	Atomic radius decreases across the period from Na to Cl (1)	1
4(c)	Nuclear charge increases / proton number increases across the period (1); Shielding remains roughly constant because electrons are added to the same principal quantum shell (n=3) (1); Greater electrostatic attraction pulls outer electrons closer to the nucleus (1)	3
5(a)	Species that possess the same number of electrons / identical electronic configuration (1)	1
5(b)	1s ² 2s ² 2p ⁶ (1)	1
5(c)	Order: N ³⁻ > O ²⁻ > F ⁻ > Na ⁺ > Mg ²⁺ (1); All have 10 electrons and identical shielding (1); Nuclear charge increases from N (7+) to Mg (12+), so electrons are pulled in more tightly as proton number increases (1)	3
6(a)	Deflected downwards towards the negative plate (1)	1
6(b)	1H ⁺ : q/m = +1/1 = 1.0 (1); 2H ⁺ : q/m = +1/2 = 0.5 (1); 4He ²⁺ : q/m = +2/4 = 0.5 (1)	3
6(c)	2H ⁺ and 4He ²⁺ will have identical deflection (1); because they possess identical charge-to-mass ratios (0.5) (1)	2
7(a)	Atoms of the same element with the same number of protons / same proton number (1); but different numbers of neutrons / different nucleon number (1)	2
7(b)	⁷⁹ / ₃₅ Br and ⁸¹ / ₃₅ Br (1 mark for correct mass numbers 79 and 81, 1 mark for atomic number 35 on both) (2)	2
7(c)	Chemical reactivity depends on electron configuration / number of outer-shell electrons (1); both isotopes have identical electron configurations [Ar]3d ¹⁰ 4s ² 4p ⁵ (1)	2
8(a)	C — Both have the identical empirical and molecular formula ratio of 2:1 (1)	1
8(b)	D ₂ O has a higher molecular mass than H ₂ O (1); leading to slightly stronger intermolecular hydrogen bonding / London dispersion forces (1)	2
9(a)	Mass-to-charge ratio of the ions (1)	1
9(b)	Ar = [(24 × 79.0) + (25 × 10.0) + (26 × 11.0)] / 100 (1); Ar = (1896.0 + 250.0 + 286.0) / 100 = 2432.0 / 100 (1); Ar = 24.32 (must be 2 decimal places) (1)	3
9(c)	Magnesium / Mg (1)	1
10(a)	m/z = 70 (from ³⁵ Cl- ³⁵ Cl), 72 (from ³⁵ Cl- ³⁷ Cl), and 74 (from ³⁷ Cl- ³⁷ Cl) (2 marks for all three, 1 mark for any two) (2)	2
10(b)	Probability 70 = 0.75 × 0.75 = 9/16 (1); Probability 72 = 2 × (0.75 × 0.25) = 6/16 (1); Probability 74 = 0.25 × 0.25 = 1/16, giving ratio 9 : 6 : 1 (1)	3
10(c)	Fragmentation of the molecular ion [Cl ₂] ⁺ into a chlorine atom and a monoatomic [³⁵ Cl] ⁺ or [³⁷ Cl] ⁺ ion (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
11(a)	Weighted average mass of an atom of an element (1); compared to 1/12th of the mass of an atom of carbon-12 (1)	2
11(b)	Carbon-12 is a common, solid element that can be handled easily and its mass is assigned exactly 12.0000 atomic mass units (1)	1
12(a)	Abundance of $^{71}\text{Ga} = (100 - x)\%$ (1)	1
12(b)	$69.72 = [69x + 71(100 - x)] / 100$ (1); $6972 = 69x + 7100 - 71x \Rightarrow 2x = 128$ (1); $x = 64.0\%$ for ^{69}Ga and 36.0% for ^{71}Ga (1)	3
13(a)	C — 10 electrons (5 orbitals $\times$ 2 electrons) (1)	1
13(b)	C — 9 orbitals (one 3s, three 3p, five 3d; total $n^2 = 9$) (1)	1
13(c)	s orbital: spherical (1); px orbital: dumb-bell / double lobe oriented along the x-axis (1)	2
14(a)	4s sub-shell has lower energy than 3d in an isolated neutral atom before filling (1)	1
14(b)	Electrons enter the lowest available energy orbital first according to the Aufbau principle (1); 4s is lower in energy than 3d so it fills first (1)	2
14(c)	Once electrons occupy 3d orbitals, the 3d sub-shell drops to a lower energy than 4s (1); 4s electrons become the outermost shell (highest principal quantum number $n=4$) and are removed first (1)	2
15(a)	$1s^2 2s^2 2p^6 3s^2 3p^4$ (1)	1
15(b)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$ (1)	1
15(c)	Expected configuration would be $3d^9 4s^2$ (1); a completely filled 3d ¹⁰ sub-shell provides enhanced symmetrical electronic stability (1)	2
15(d)	[Ar] $3d^5$ (1 mark for removing two 4s electrons first, 1 mark for removing one 3d electron leaving $3d^5$) (2)	2
16(a)	Orbitals of equal energy (degenerate orbitals) are each occupied singly with parallel spins (1); before any orbital is doubly occupied / paired (1)	2
16(b)	Three degenerate 2p orbital boxes drawn (1); each box contains exactly one single arrow pointing in the same direction (three unpaired electrons) (1)	2
16(c)	Five 3d orbital boxes, each containing one single arrow with parallel spin (1); 5 unpaired electrons (1)	2
17(a)	B — $\text{Cl}\cdot$ has 7 valence electrons, with 1 unpaired electron (1)	1
17(b)	3 unpaired electrons (in the 3px, 3py, 3pz orbitals) (1)	1
17(c)	Cl^- has a full octet / fully filled 3p sub-shell with all electrons paired (noble gas configuration) (1); $\text{Cl}\cdot$ has an incomplete valence shell and an unpaired electron seeking a pair (1)	2
18(a)	$1s^2 2s^2 2p^6 3s^2 3p^4$ (1)	1
18(b)	Group 16 (or VI) (1); Period 3 (1)	2
18(c)	Sulfur / S (1)	1
19(a)	Energy required to remove one electron from each atom (1); in one mole of gaseous atoms (1); to form one mole of gaseous $1+$ ions (1)	3
19(b)	$\text{Mg}^+(\text{g}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{e}^-$ (1 mark for correct formulae and state symbols, 1 mark for balanced electron) (2)	2
20(a)	General trend increases across Period 3 (1); Nuclear charge increases as proton number increases while shielding remains similar (same shell $n=3$) (1); Greater electrostatic attraction holds valence electrons more tightly (1)	3
20(b)	Al electron removed from 3p sub-shell; Mg electron removed from 3s sub-shell (1); 3p sub-shell is higher in energy / further from the nucleus than 3s (1); 3p electron experiences additional shielding from the filled 3s ² electrons (1)	3

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
20(c)	In phosphorus, all three 3p electrons occupy separate orbitals singly (3p1 3p1 3p1) (1); in sulfur, two electrons are paired in one 3p orbital (3p2 3p1 3p1) (1); mutual spin repulsion between the paired electrons in sulfur lowers the energy required to remove one (1)	3
21(a)	Removing successive electrons leaves a progressively more positive ion with the same nuclear charge (1); the remaining electrons experience greater electrostatic attraction per electron and are pulled closer to the nucleus (1)	2
21(b)	Large jump between 5th and 6th electrons (1); very large jump between 13th and 14th electrons (1)	2
21(c)	Group 15 (or Group 5) (1); the first 5 electrons are relatively easy to remove before a large jump to the 6th electron, showing 5 valence electrons (1)	2
21(d)	Phosphorus / P (15 total electrons) (1)	1
22(a)	Between the 3rd and 4th ionisation energies (11578 / 2745 = 4.2× jump) (1)	1
22(b)	3 valence electrons (1)	1
22(c)	YCl3 (or AlCl3) (1)	1
23(a)	First ionisation energy decreases down Group 2 (1)	1
23(b)	Down the group, the number of occupied electron shells increases / atomic radius increases (1); outer electrons experience increased shielding from extra inner electron shells (1); the increased shielding and distance outweigh the increase in nuclear charge, so outer electrons are held less tightly (1)	3
24(a)	Partially true / False because there are two distinct exceptions (1); General trend increases due to increasing nuclear charge with constant shielding (1); Exception 1: Al (578) is lower than Mg (738) (1); because Al removes electron from 3p sub-shell which is higher in energy / shielded by 3s2 (1); Exception 2: S (1000) is lower than P (1012) (1); because S has a paired electron in 3p (3p4 vs 3p3 in P), resulting in spin-pair repulsion that facilitates electron removal (1)	6
25(a)	Path deflected towards positive plate: electrons (1); Straight path: neutrons (1); Path deflected towards negative plate: protons (1)	3
25(b)	Deflection angle is proportional to charge/mass (q/m) ratio (1); Electrons have a mass roughly 1/1840 of a proton, giving a vastly higher q/m ratio and greater acceleration (1)	2
25(c)	Charge on the particle / mass of the particle (or charge-to-mass ratio) (1); Strength of electric field / velocity (speed) of the particle (1)	2
26(a)	Protons = 16 (1); Neutrons = 34 - 16 = 18 (1); Electrons = 16 + 2 = 18 (1)	3
26(b)	Protons = 26 (1); Neutrons = 56 - 26 = 30 (1); Electrons = 26 - 3 = 23 (1)	3
26(c)	39K+ (39-19=20) and 40Ca2+ (40-20=20) (1)	1
27(a)	Atomic radius increases down the group (1); Number of occupied principal electron shells increases (1); Outermost electrons are further from the nucleus and experience increased inner-shell shielding (1)	3
27(b)	Br- gains an electron into the 4p sub-shell, increasing electron-electron repulsion (1); The nuclear charge (+35) remains unchanged, allowing the electron cloud to expand (1)	2
28(a)	$q/m = 1.60 \times 10^1 / 1.67 \times 10^2$ (1) = 9.58×10 C kg ⁻¹ (1)	2
28(b)	$q = 2 \times (1.60 \times 10^1) = 3.20 \times 10^1$ C, $m = 4 \times (1.67 \times 10^2) = 6.68 \times 10^2$ kg (1); $q/m = 4.79 \times 10$ C kg ⁻¹ (1)	2
28(c)	Approximately 1836 to 1840 : 1 (1)	1
29(a)	All Period 3 elements possess the identical core electron configuration 1s2 2s2 2p6 (10 core electrons) (1); Incoming valence electrons are added to the same outermost principal quantum shell (n=3) (1)	2
29(b)	Nuclear charge increases by +1 with each successive element (from +11 to +17) (1); Because core shielding is constant, Z _{eff} increases steadily across the period (1); Greater Z _{eff} pulls the valence shell closer to the nucleus, causing atomic radius to decrease (1)	3
30(a)	Most alpha particles passed straight through the gold foil with little or no deflection (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
30(b)	A small fraction of alpha particles were deflected through very large angles ($> 90^\circ$) / bounced straight back (1); Showing that the positive charge and mass are concentrated in a tiny central nucleus (1)	2
31(a)	5 isotopes (1)	1
31(b)	$Ar = [(90 \times 51.5) + (91 \times 11.2) + (92 \times 17.1) + (94 \times 17.4) + (96 \times 2.8)] / 100$ (1); $Ar = [4635.0 + 1019.2 + 1573.2 + 1635.6 + 268.8] / 100 = 9131.8 / 100$ (1); $Ar = 91.32$ (must be 2 d.p.) (1)	3
31(c)	Most abundant is ^{90}Zr ; neutrons = $90 - 40 = 50$ (1)	1
32(a)	$B - 10B = 19.0\%$, $11B = 81.0\%$ (1)	1
32(b)	$10.81 = [10x + 11(100 - x)] / 100$ (1); $1081 = 1100 - x \Rightarrow x = 19.0\%$ (1)	2
33(a)	Mass of an atom of an isotope (1); relative to $1/12$ th of the mass of an atom of carbon-12 (1)	2
33(b)	$Ar = [(204 \times 1.4) + (206 \times 24.1) + (207 \times 22.1) + (208 \times 52.4)] / 100$ (1); $Ar = [285.6 + 4964.6 + 4574.7 + 10899.2] / 100 = 20724.1 / 100$ (1); $Ar = 207.2$ (1)	3
34(a)	$m/z = 158$ ($^{79}\text{Br}-^{79}\text{Br}$), $m/z = 160$ ($^{79}\text{Br}-^{81}\text{Br}$ and $^{81}\text{Br}-^{79}\text{Br}$), $m/z = 162$ ($^{81}\text{Br}-^{81}\text{Br}$) (2 marks for all three, 1 mark for any two) (2)	2
34(b)	$\text{Prob}(158) = 0.5 \times 0.5 = 0.25$ (1); $\text{Prob}(160) = 2 \times (0.5 \times 0.5) = 0.50$ (1); $\text{Prob}(162) = 0.5 \times 0.5 = 0.25$, giving ratio 1 : 2 : 1 (1)	3
35(a)	Exactly 12 unified atomic mass units (or 12 u / 12 Da) (1)	1
35(b)	Mass number is the sum of whole protons and neutrons in a single nucleus (integers) (1); Relative atomic mass is a weighted average of isotopic masses based on natural abundance (1)	2
36(a)	ionisation, acceleration, ion drift / flight tube, detection (2 marks for all four, 1 mark for any two/three) (2)	2
36(b)	Ions are accelerated by an electric potential difference (V) so that $KE = 1/2 mv^2 = qV$ (1); With constant KE, velocity depends solely on the mass of the ion (1)	2
36(c)	Lighter ions (lower m/z) travel faster and arrive at the detector in a shorter time (1); Flight time is proportional to $\sqrt{(m/z)}$ (1)	2
37(a)	$28.09 = [(28 \times 92.2) + (29 \times 4.7) + (A \times 3.1)] / 100$ (1); $2809 = 2581.6 + 136.3 + 3.1A \Rightarrow 3.1A = 91.1$ (1); $A = 91.1 / 3.1 = 29.39 \Rightarrow$ mass number $A = 30$ (1)	3
37(b)	Neutrons = $30 - 14 = 16$ (1)	1
38(a)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$ (1)	1
38(b)	A half-filled 3d sub-shell possesses extra exchange energy / symmetrical stability (1); Symmetrical distribution of spin-aligned electrons minimizes inter-electronic repulsion (1)	2
38(c)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$ (4s electron is removed first) (1)	1
39(a)	$[\text{Ar}] 3d^7$ (1)	1
39(b)	$[\text{Ar}] 3d^8$ (1)	1
39(c)	Once electrons occupy 3d orbitals, they shield the 4s electrons (1); 3d orbitals drop to a lower energy level than 4s, making 4s electrons the outermost valence electrons (1)	2
40(a)	C — 32 electrons ($2n^2 = 2 \times 4^2 = 32$) (1)	1
40(b)	4s (1 orbital), 4p (3 orbitals), 4d (5 orbitals), 4f (7 orbitals) (2 marks for all four correct, 1 mark for any two/three) (2)	2
41(a)	Orbitals that possess identical energy levels within an isolated atom (1)	1
41(b)	Dumbbell-shaped lobes oriented mutually perpendicular (at 90° angles) to one another along the x, y, and z Cartesian axes (1); Each with an electron nodal plane at the nucleus (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
41(c)	A 2s orbital is larger in size / has greater radial extent than 1s (1); A 2s orbital has higher energy than 1s (or 2s has a radial node) (1)	2
42(a)	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ (1)	1
42(b)	Argon / Ar (1)	1
42(c)	1s ² 2s ² 2p ⁶ (1)	1
43(a)	Opposite spins generate opposing magnetic fields (1); This partially cancels the electrostatic repulsion between two negatively charged electrons in the same space (1)	2
43(b)	Three 2p boxes drawn (1); One box contains two paired opposite-spin arrows; the remaining two boxes each contain one single parallel arrow (1); Resulting in 1 paired orbital and 2 unpaired electrons (1)	3
44(a)	Both periods show a general increase from Group 1 to Group 18 (1); Both display two dips: between Groups 2 and 13, and between Groups 15 and 16 (1)	2
44(b)	Period 3 elements have their outermost electron in the n=3 shell whereas Period 2 is in n=2 (1); The outer electron is further from the nucleus (larger atomic radius) (1); Greater shielding from inner core electron shells outweighs the increased nuclear charge (1)	3
44(c)	Between Be and B in Period 2 / Mg and Al in Period 3 (1); Between N and O in Period 2 / P and S in Period 3 (1)	2
45(a)	Nitrogen: 2p _x ¹ 2p _y ¹ 2p _z ¹ (1); Oxygen: 2p _x ² 2p _y ¹ 2p _z ¹ (1)	2
45(b)	In oxygen, two electrons must pair up in one 2p orbital (1); The mutual electrostatic repulsion between the two paired electrons in the same orbital destabilizes them (1); This spin-pair repulsion makes one electron easier to remove despite the higher nuclear charge of oxygen (1)	3
46(a)	Between the 1st and 2nd ionisation energies (4563 vs 496 kJ mol ⁻¹) (1)	1
46(b)	The 1st electron is removed from the outer 3s shell; the 2nd electron is removed from the inner, complete 2p shell (1); The 2p electron is much closer to the nucleus and experiences much less shielding (1)	2
46(c)	The 10th and 11th electrons are removed from the innermost 1s principal quantum shell (n=1) (1); They are closest to the nucleus with zero inner shielding, experiencing maximum nuclear attraction (1)	2
47(a)	Group 14 (or Group 4) (1); There is a massive jump between the 4th and 5th ionisation energies (6223 to 37832), indicating 4 valence electrons before breaking into an inner shell (1)	2
47(b)	Carbon / C (Period 2, Group 14) (1)	1
47(c)	CO ₂ (1)	1
48(a)	B — Increased atomic radius and shielding by inner electron shells (1)	1
48(b)	Each row down the group adds a completely new principal quantum shell of core electrons (1); The shielding effect and greater distance of the valence electron from the nucleus outweigh the added nuclear charge (1)	2
49(a)	Al(g) → Al ⁺ (g) + e ⁻ (1)	1
49(b)	Al ²⁺ (g) → Al ³⁺ (g) + e ⁻ (1 mark for formulae and charge, 1 mark for (g) state symbols) (2)	2
49(c)	The 3rd electron is removed from a dipositive cation (Al ²⁺) with +13 nuclear charge attracting only 11 remaining electrons (1); Greater effective attraction and smaller ionic radius require much more energy (1)	2
50(a)	Group 2 (large jump between 2nd and 3rd IE) (1); Electron configuration: 1s ² 2s ² 2p ⁶ 3s ² (1)	2
50(b)	Protons = 12 (1); Neutrons = 24 - 12 = 12 (1); Electrons = 12 - 2 = 10 (1)	3
50(c)	The first two electrons are removed from the outer 3s sub-shell (n=3) (1); The 3rd electron is removed from the inner, complete 2p sub-shell (n=2) (1); The 2p sub-shell is much closer to the nucleus, has significantly less shielding, and experiences an immensely stronger electrostatic pull (1)	3